

# Personal Website Development Plan

## Vision

Build a personal website that feels less like a portfolio and more like an interactive exploration of my work, research, and interests.

The website should be:

- Visually distinctive and memorable
- Professional enough for professors, researchers, recruiters, and collaborators
- Exploration-driven rather than résumé-driven
- Clean and modern rather than flashy
- Focused on projects, ideas, and research questions

The goal is that visitors remember the website itself, not just the contents.

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## Core Concept

The homepage is an interactive map.

The map is the primary navigation system for the entire website.

Unlike most portfolio websites, users do not begin by scrolling through sections. Instead, they explore the map and discover projects through interaction.

The map represents my intellectual landscape rather than geographic locations.

Projects appear as landmarks across the map.

Related projects are positioned near one another.

Connections between projects can be represented visually through roads, rivers, paths, or proximity.

The map exists to organize information spatially, not metaphorically.

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## Important Design Principle

### No Corny Naming

Avoid names such as:

- Hilbert Mountains
- Encryption Desert

- Kingdom of Biology
- Forest of Algorithms

Projects should be labeled with their actual names.

Examples:

- Long Inverted Repeat Detection
- Pareto-Optimal Polymer Enumeration
- Privacy-Preserving Genomics
- RNA Interaction Energy Distribution
- Game Boy Emulator
- Medical Device Startup

The map aesthetic is fantasy-inspired.

The project naming is not.

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## Visual Style

### Inspiration

The visual design should draw from:

- Fantasy cartography
- Exploration maps
- Hand-drawn atlases
- Scientific illustrations
- Old expedition maps

The design should NOT resemble:

- RPG games
  - MMORPG maps
  - Video game UIs
  - Dungeons & Dragons interfaces
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### Desired Feel

Keywords:

- Elegant
- Exploratory
- Sophisticated
- Curious
- Technical
- Minimal

Visitors should feel:

"This is an interesting researcher."

not

"This person built a game."

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## Homepage Experience

Visitors land directly on the world map.

The map fills most of the screen.

Visible immediately:

- Project landmarks
- Terrain
- Connections
- Points of interest

Hovering over a project reveals:

- Project title
- Short description
- Status
- Tags

Example:

Long Inverted Repeat Detection

Detecting large imperfect inverted repeats with long spacers.

Status: Active

Tags: Computational Biology Algorithms Genomics

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Clicking a project zooms into a dedicated project page.

The transition should feel similar to:

- Google Maps
  - Geographic zooming
  - Camera navigation
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# Information Architecture

## Homepage

Interactive map

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## Project Pages

Each project should contain:

### Problem

What problem is being solved?

### Motivation

Why does the problem matter?

### Approach

Methods and techniques used.

### Results

Key findings and outcomes.

### Artifacts

Links to:

- Papers
- Repositories
- Posters
- Presentations

### Timeline

Project history and milestones.

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## Research Page

Optional aggregate view of research projects.

Provides:

- Publications

- Research interests
  - Collaborators
  - Current work
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## Writing / Notes

Collection of:

- Research notes
- Paper reviews
- Technical writeups
- Explorations

Potential topics:

- Long inverted repeats
  - Hilbert basis computation
  - Homomorphic encryption
  - Computational biology
  - Algorithm design
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## CV Page

Traditional academic CV.

Should remain simple and professional.

PDF download available.

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## Map Structure

The map is not divided by academic disciplines.

Instead, projects naturally occupy locations.

The geography exists primarily to:

- Encourage exploration
- Create memorable navigation
- Visually represent relationships

Potential structures:

- Roads between related projects
- Rivers connecting research themes
- Terrain separating unrelated work

- Coastlines defining major domains
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## Easter Eggs

Easter eggs should feel like discoveries.

Avoid arbitrary key combinations.

Avoid cheat-code style interactions.

Good examples:

## Hidden Cabin

Opens:

Conway's Game of Life simulation

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## Cave

Contains:

Experimental projects

Failed ideas

Abandoned research directions

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## Telescope

Displays:

Future project ideas

Research questions

Open problems

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## Secret Area

Contains:

Algorithm visualizations

Interactive simulations

Random technical experiments

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## Technical Stack

### Frontend

- Next.js
  - TypeScript
  - Tailwind CSS
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### Animation

- Framer Motion
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### Content

- MDX
  - Content Collections
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### Hosting

- Vercel
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### Domain

Future purchase:

Preferred options:

- architpatil.com
- archit.dev
- architpatil.me

Domain purchase is not required for initial development.

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# Development Roadmap

## Phase 1 — Foundation

Goal:

Establish project structure.

Tasks:

- Initialize Next.js project
- Configure Tailwind
- Configure MDX
- Deploy initial Vercel version
- Create project content schema

Deliverable:

Basic website framework.

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## Phase 2 — Map Prototype

Goal:

Create first interactive map.

Tasks:

- Draw map layout
- Add project markers
- Implement hover cards
- Implement zoom behavior

Deliverable:

Fully navigable map homepage.

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## Phase 3 — Project Pages

Goal:

Build project content system.

Tasks:

- Project template



- MDX integration
- Project metadata
- Research pages

Deliverable:

All major projects accessible.

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## Phase 4 — Visual Polish

Goal:

Create memorable experience.

Tasks:

- Terrain improvements
- Smooth transitions
- Hover animations
- Loading states
- Typography improvements

Deliverable:

Professional visual identity.

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## Phase 5 — Easter Eggs

Goal:

Add hidden content.

Tasks:

- Conway's Game of Life
- Secret locations
- Interactive experiments

Deliverable:

Discoverable hidden features.

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## Phase 6 — Public Launch

Tasks:

- Purchase domain
- Connect domain
- SEO
- Analytics
- Final testing

Deliverable:

Production-ready website.

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## Success Criteria

The website succeeds if visitors:

1. Immediately recognize it as different from a standard portfolio.
2. Can understand my research and projects quickly.
3. Remember the website after leaving.
4. Explore multiple projects voluntarily.
5. View me as someone actively investigating difficult technical problems rather than simply listing accomplishments.

The map should serve the content.

The content should never be sacrificed for the map.